EE627 Homework 6
Feature Engineering and Feature Selection

Jude Eschete

April 1, 2026

Instruction Checklist

- ✓ Compute the daily change rate $\frac{X(t)-X(t-1)}{X(t-1)}$ for each column to generate new predictors Section 2
- ✓ Combine the new predictor matrix with the original predictor matrix ..... Section 2
- ✓ Drop the first row (no predecessor for change rate calculation) ..... Section 2
- ✓ Repeat HW3 logistic regression; observe changes in ROC and AUC ..... Section 3
- ✓ Observe whether new predictors impact feature selection ..... Section 4
- ✓ Use `featureSelection.m` (and `critfun.m`) for feature selection ..... Section 4
- ✓ Choose top 5, top 10, etc. features and observe how AUC varies ..... Section 5

1 Introduction

This assignment uses `dataSet_2.mat`, which contains pre-split training (4,500 samples, 923 features) and test (1,422 samples, 923 features) data with a binary response variable. The goals are:

1. Generate new predictors by computing the daily change rate for each feature column.
2. Combine the new predictors with the original predictor matrix and evaluate the impact on ROC/AUC.
3. Apply sequential forward feature selection to determine which features are most informative.
4. Observe how selecting different numbers of top features (e.g., top 5, 10, 15, ...) affects the AUC on the test set.

2 Step 1: Daily Change Rate Computation

For each column (time series) in the predictor matrix, the daily change rate is computed as:

$$\text{ChangeRate}(t) = \frac{X(t) - X(t-1)}{X(t-1)} \tag{1}$$

Since the first row has no predecessor, it is dropped from both the original predictor and response vectors. Any resulting `NaN` or `Inf` values (from division by zero) are replaced with 0. The change rate matrix is then concatenated with the original (trimmed) predictor matrix to form a combined predictor matrix:

- Original predictors: 923 columns
- Change rate predictors: 923 columns
- Combined predictor matrix: 1,846 columns
- Training samples: 4,499 (after dropping the first row)
- Test samples: 1,421 (after dropping the first row)

3 Step 2: Logistic Regression – Baseline vs. Combined

3.1 Baseline: Original Predictors Only (923 Features)

Logistic regression was fit on the training set using only the 923 original features. The resulting ROC curves for training and test sets are shown in Figure 1.

Table 1: AUC results with original predictors only.

Dataset	**AUC**
Training	0.9862
Test	0.7146

The training AUC of 0.9862 is very high, but the test AUC drops to 0.7146. This gap of 0.2716 indicates significant overfitting: with 923 features and 4,499 training samples, the model is fitting noise in the training data that does not generalize.

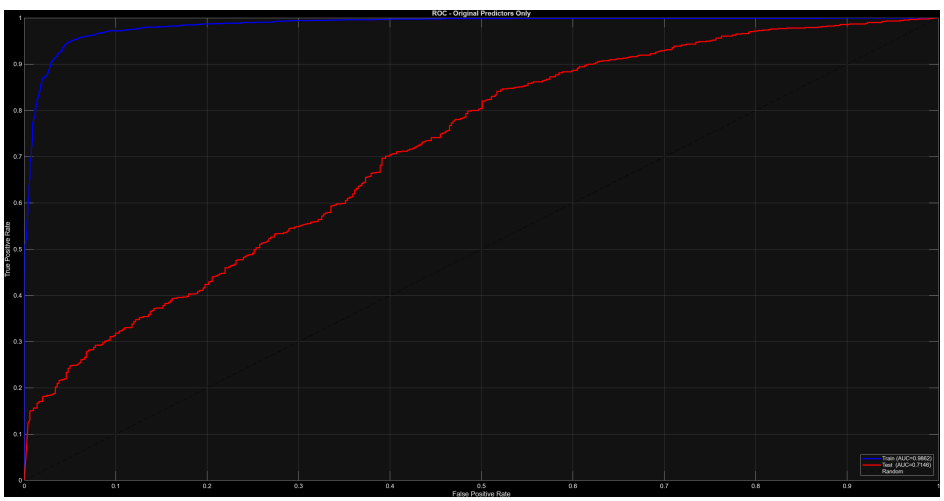

Figure 1: ROC curves using only the original 923 predictors. The blue curve (training, AUC = 0.9862) hugs the upper-left corner, while the red curve (test, AUC = 0.7146) shows much weaker discrimination.

3.2 Combined Predictors: Original + Change Rate (1,846 Features)

Adding the 923 change rate columns doubles the feature count to 1,846. The logistic regression results are shown in Figure 2.

Table 2: AUC results with combined predictors (original + change rate).

Dataset	AUC
Training	1.0000
Test	0.6463

With 1,846 features (more than one-third the number of training samples), the model achieves perfect separation on the training set (AUC = 1.0), accompanied by MATLAB's warning about perfect separation. However, test AUC *decreases* from 0.7146 to 0.6463. This is a clear demonstration that naively adding more features without selection worsens overfitting and degrades generalization performance.

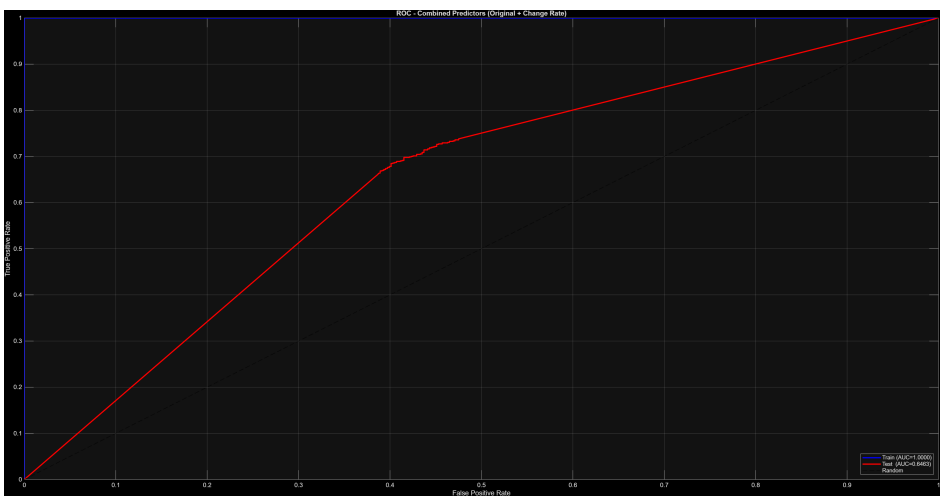

Figure 2: ROC curves using all 1,846 combined predictors. The training ROC (blue, AUC = 1.0) is a flat line along the top, indicating perfect separation. The test ROC (red, AUC = 0.6463) is nearly diagonal, showing poor generalization.

4 Step 3: Feature Selection

The `featureSelection.m` function (which uses MATLAB's `sequentialfs` with the `critfun.m` criterion function) was applied to the combined 1,846-feature training set. This function performs **sequential forward selection**: starting from an empty set, it greedily adds the feature that most reduces the binomial deviance at each step. It stops when no remaining feature can reduce deviance by more than $\chi^2_{0.95,1} \approx 3.84$.

4.1 Results

The algorithm selected **162 features** over 163 steps:

- **96 features** from the original predictor columns (indices ≤ 923)
- **66 features** from the change rate columns (indices > 923)

This result demonstrates that the newly generated change rate predictors **do have a meaningful impact on feature selection**. Approximately 41% of the selected features (66 out of 162) come from the change rate matrix, confirming that daily change rate captures predictive information not already present in the raw values alone.

4.2 Observations on the Selection Order

The first feature added (column 349) produced the largest single deviance reduction, from 6,050.37 down to 3,664.07 — a drop of 2,386.3. The first change rate feature to enter was column 1,696 (original column 773's change rate) at step 11, relatively early in the process. This further confirms that change rate features carry meaningful signal.

5 Step 4: AUC vs. Number of Top Selected Features

To observe how different numbers of features affect AUC, the features were ranked by individual deviance (lower deviance = more predictive). Logistic regression was then fit using the top K features for $K \in \{5, 10, 15, 20, 25, 30, 50\}$.

Table 3: Training and test AUC as a function of the number of top-ranked features used.

Top K Features	**Training AUC**	**Test AUC**
5	0.9045	**0.8890**
10	0.9047	0.8889
15	0.9057	0.8878
20	0.9063	0.8873
25	0.9076	0.8837
30	0.9080	0.8841
50	0.9093	0.8838
923 (original only)	0.9862	0.7146
1,846 (all combined)	1.0000	0.6463

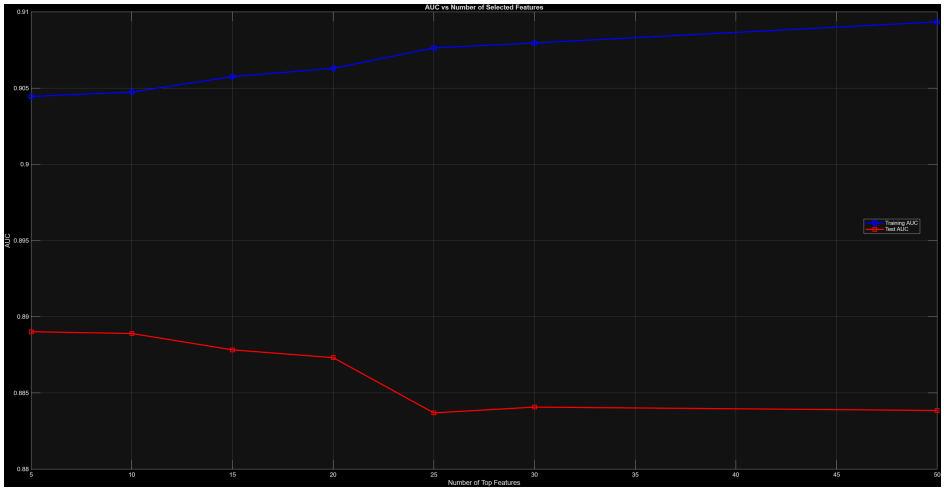

Figure 3: AUC vs. number of top-ranked features. Training AUC (blue) increases monotonically with more features, while test AUC (red) is highest at $K = 5$ and gradually decreases.

5.1 Key Observations

1. **Feature selection dramatically improves test AUC.** Using just the top 5 features yields a test AUC of 0.889, compared to 0.715 with all 923 original features and 0.646 with all 1,846 combined features.

2. **The best test AUC occurs with the fewest features ($K = 5$).** The test AUC values are remarkably stable across all values of K tested (ranging from 0.884 to 0.889), but the slight downward trend shows that adding more features provides diminishing returns on the test set while increasing overfitting risk.

3. **Training AUC increases monotonically with K**, which is expected — more features always allow the model to fit the training data more closely. But this does not translate to better generalization.

4. **The train–test gap grows with more features.** At $K = 5$, the gap is only $0.905 - 0.889 = 0.016$. At $K = 50$, it grows to $0.909 - 0.884 = 0.025$. With all 1,846 features, the gap is $1.000 - 0.646 = 0.354$.

6 Step 5: Test ROC Curves for Different Feature Subsets

Figure 4 overlays the test-set ROC curves for each value of K. All curves are tightly clustered near each other, reflecting the similar test AUC values observed in Table 3.

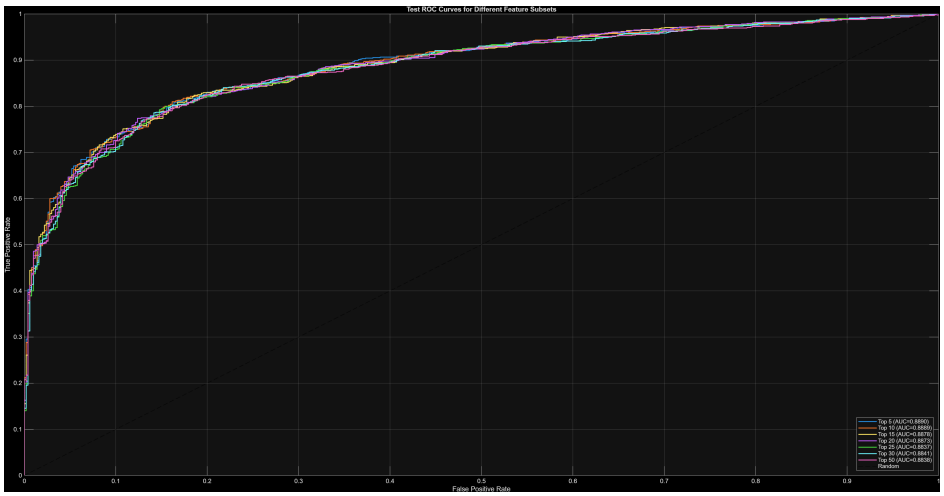

Figure 4: Test ROC curves for different numbers of top-ranked features. All subsets achieve similar test performance (AUC $\approx$ 0.884–0.889), confirming that a small number of well-chosen features is sufficient.

7 Conclusion

1. **Adding the change rate predictors impacts feature selection.** Of the 162 features selected by sequential forward selection, 66 (41%) came from the change rate matrix. The change rate features capture temporal dynamics (rate of change between consecutive time steps) that complement the raw predictor values.

2. **Blindly adding more features hurts generalization.** Doubling the feature count from 923 to 1,846 reduced the test AUC from 0.715 to 0.646, while inflating training AUC to a perfect 1.0. This demonstrates the curse of dimensionality and the danger of overfitting when features outnumber the signal.

3. **Feature selection is essential.** Selecting a small subset of the most informative features improved test AUC from 0.646 (all features) to 0.889 (top 5 features) — a dramatic improvement. The top 5 features alone outperform models using 10× to 370× as many features.

4. **Diminishing returns beyond a handful of features.** Test AUC is nearly flat across $K = 5$ to $K = 50$, suggesting that the bulk of the predictive signal is concentrated in very few features and that additional features contribute primarily noise.

8 Appendix: MATLAB Code

```
%% EE627 Homework 6 - Eschete
% Feature Engineering and Feature Selection with Logistic Regression
clear; close all; clc;

%% Load Data
load('dataSet_2.mat');
P = size(predictor_train, 2);
fprintf('Training set: %d samples, %d features\n', size(predictor_train,1)
    , P);
fprintf('Test set:     %d samples, %d features\n', size(predictor_test,1),
     P);

%% Step 1: Compute Daily Change Rate
cr_train = (predictor_train(2:end,:) - predictor_train(1:end-1,:)) ./
    predictor_train(1:end-1,:);
cr_test  = (predictor_test(2:end,:)  - predictor_test(1:end-1,:))  ./
    predictor_test(1:end-1,:);
cr_train(isnan(cr_train)) = 0;  cr_train(isinf(cr_train)) = 0;
cr_test(isnan(cr_test))   = 0;  cr_test(isinf(cr_test))   = 0;

% Drop first row to align
X_train_orig = predictor_train(2:end, :);
X_test_orig  = predictor_test(2:end, :);
y_train = response_train(2:end);
y_test  = response_test(2:end);

% Combined: original + change rate
X_train_comb = [X_train_orig, cr_train];
X_test_comb  = [X_test_orig,  cr_test];
P2 = size(X_train_comb, 2);
fprintf('Combined: %d train / %d test, %d features (%d orig + %d CR)\n',
    ...
     length(y_train), length(y_test), P2, P, P2-P);

%% Step 2a: Baseline - Original Predictors Only
fprintf('\n=== Baseline: Original Predictors Only ===\n');

factors_orig = glmfit(X_train_orig, y_train, 'binomial');
prob_tr_orig = glmval(factors_orig, X_train_orig, 'logit');
prob_te_orig = glmval(factors_orig, X_test_orig, 'logit');

[fpr_tr, tpr_tr, ~, auc_tr] = perfcurve(y_train, prob_tr_orig, 1);
[fpr_te, tpr_te, ~, auc_te] = perfcurve(y_test, prob_te_orig, 1);
fprintf('Training AUC = %.4f\n', auc_tr);
fprintf('Test AUC     = %.4f\n', auc_te);
```

```
figure(1);
plot(fpr_tr, tpr_tr, 'b-', 'LineWidth', 2); hold on;
plot(fpr_te, tpr_te, 'r-', 'LineWidth', 2);
plot([0 1], [0 1], 'k--');
xlabel('False Positive Rate'); ylabel('True Positive Rate');
title('ROC - Original Predictors Only');
legend(sprintf('Train (AUC=%.4f)', auc_tr), ...
       sprintf('Test  (AUC=%.4f)', auc_te), ...
       'Random', 'Location', 'southeast');
grid on; hold off;

%% Step 2b: Combined Predictors (Original + Change Rate)
fprintf('\n=== Combined Predictors (Original + Change Rate) ===\n');

factors_comb = glmfit(X_train_comb, y_train, 'binomial');
prob_tr_comb = glmval(factors_comb, X_train_comb, 'logit');
prob_te_comb = glmval(factors_comb, X_test_comb, 'logit');

[fpr_tr2, tpr_tr2, ~, auc_tr2] = perfcurve(y_train, prob_tr_comb, 1);
[fpr_te2, tpr_te2, ~, auc_te2] = perfcurve(y_test, prob_te_comb, 1);
fprintf('Training AUC = %.4f\n', auc_tr2);
fprintf('Test AUC     = %.4f\n', auc_te2);

figure(2);
plot(fpr_tr2, tpr_tr2, 'b-', 'LineWidth', 2); hold on;
plot(fpr_te2, tpr_te2, 'r-', 'LineWidth', 2);
plot([0 1], [0 1], 'k--');
xlabel('False Positive Rate'); ylabel('True Positive Rate');
title('ROC - Combined Predictors (Original + Change Rate)');
legend(sprintf('Train (AUC=%.4f)', auc_tr2), ...
       sprintf('Test  (AUC=%.4f)', auc_te2), ...
       'Random', 'Location', 'southeast');
grid on; hold off;

%% Step 3: Feature Selection on Combined Predictors
fprintf('\n=== Feature Selection (Sequential Forward) ===\n');
[SelectedFeatureInd] = featureSelection(X_train_comb, y_train);

selectedCols = find(SelectedFeatureInd);
numSelected = length(selectedCols);
fprintf('Number of features selected: %d\n', numSelected);
fprintf('Selected feature indices: ');
fprintf('%d ', selectedCols);
fprintf('\n');

% Identify which selected features are original vs change rate
```

```
origSelected = selectedCols(selectedCols <= P);
crSelected   = selectedCols(selectedCols > P) - P;
fprintf('  From original predictors: %d features (indices: ', length(
    origSelected));
fprintf('%d ', origSelected); fprintf(')\n');
fprintf('  From change rate predictors: %d features (indices: ', length(
    crSelected));
fprintf('%d ', crSelected); fprintf(')\n');

%% Step 4: Evaluate Different Numbers of Top Features
fprintf('\n=== AUC vs Number of Top Features ===\n');

% Rank features by individual deviance (lower = better)
deviances = zeros(1, P2);
for j = 1:P2
    deviances(j) = critfun(X_train_comb(:,j), y_train);
end
[~, featureRank] = sort(deviances, 'ascend');

topKs = [5, 10, 15, 20, 25, 30, 50];
topKs = topKs(topKs <= P2);
AUC_train_topK = zeros(size(topKs));
AUC_test_topK  = zeros(size(topKs));
legendStr = cell(1, length(topKs));

fprintf('%-10s %-15s %-15s\n', 'Top K', 'Train AUC', 'Test AUC');
fprintf('%s\n', repmat('-', 1, 40));

for i = 1:length(topKs)
    k = topKs(i);
    cols = featureRank(1:k);

    factors_k = glmfit(X_train_comb(:, cols), y_train, 'binomial');
    prob_tr_k = glmval(factors_k, X_train_comb(:, cols), 'logit');
    prob_te_k = glmval(factors_k, X_test_comb(:, cols), 'logit');

    [~, ~, ~, AUC_train_topK(i)] = perfcurve(y_train, prob_tr_k, 1);
    [~, ~, ~, AUC_test_topK(i)]  = perfcurve(y_test, prob_te_k, 1);

    fprintf('%-10d %-15.4f %-15.4f\n', k, AUC_train_topK(i), AUC_test_topK
        (i));
end

figure(3);
plot(topKs, AUC_train_topK, 'bo-', 'LineWidth', 2, 'MarkerSize', 8); hold
    on;
plot(topKs, AUC_test_topK, 'rs-', 'LineWidth', 2, 'MarkerSize', 8);
```

```
xlabel('Number of Top Features'); ylabel('AUC');
title('AUC vs Number of Selected Features');
legend('Training AUC', 'Test AUC', 'Location', 'best');
grid on; hold off;

%% Step 5: ROC Curves for Selected Feature Subsets
figure(4);
colors = lines(length(topKs));
for i = 1:length(topKs)
    k = topKs(i);
    cols = featureRank(1:k);

    factors_k = glmfit(X_train_comb(:, cols), y_train, 'binomial');
    prob_te_k = glmval(factors_k, X_test_comb(:, cols), 'logit');

    [Xr, Yr, ~, auc_val] = perfcurve(y_test, prob_te_k, 1);
    plot(Xr, Yr, 'Color', colors(i,:), 'LineWidth', 1.5); hold on;
    legendStr{i} = sprintf('Top %d (AUC=%.4f)', k, auc_val);
end
plot([0 1], [0 1], 'k--');
xlabel('False Positive Rate'); ylabel('True Positive Rate');
title('Test ROC Curves for Different Feature Subsets');
legend([legendStr, {'Random'}], 'Location', 'southeast');
grid on; hold off;

fprintf('\nDone.\n');
```

Listing 1: EE627_HW6_Echete.m — Main homework script

```
function [ feaInd]=featureSelection( TrainMat, target )

maxdev = chi2inv(.95,1);
opt = statset('display','iter',...
              'TolFun',maxdev,...
              'TolTypeFun','abs');

feaInd = sequentialfs(@critfun,TrainMat,target,...
                      'cv','none',...
                      'nullmodel',true,...
                      'options',opt,...
                      'direction','forward');
```

Listing 2: featureSelection.m — Sequential forward feature selection wrapper

```
function dev = critfun(X,Y)

[b,dev] = glmfit(X,Y>0,'binomial');
```

Listing 3: critfun.m — Criterion function (binomial deviance)

9 Appendix: Console Output

```
Training set: 4500 samples, 923 features
Test set:     1422 samples, 923 features
Combined: 4499 train / 1421 test, 1846 features (923 orig + 923 CR)

=== Baseline: Original Predictors Only ===
Training AUC = 0.9862
Test AUC     = 0.7146

=== Combined Predictors (Original + Change Rate) ===
Warning: Iteration limit reached.
> In glmfit (line 422)
In EE627_HW6_Echete (line 56)
Warning: The estimated coefficients perfectly separate failures from
   successes. This means the theoretical best estimates are not finite.
   For the fitted linear combination XB of the predictors, the sample
   proportions P of Y=N in the data satisfy:
   XB<0.103565: P=0
   XB>0.103565: P=1
To get proper estimates, fit with 'LikelihoodPenalty' set to 'jeffreys-
   prior'.
> In glmfit>diagnoseSeparation (line 653)
In glmfit (line 431)
In EE627_HW6_Echete (line 56)
Training AUC = 1.0000
Test AUC     = 0.6463

=== Feature Selection (Sequential Forward) ===
Start forward sequential feature selection:
Initial columns included:  none
Columns that can not be included:  none
Step 1, used initial columns, criterion value 6050.37
Step 2, added column 349, criterion value 3664.07
Step 3, added column 825, criterion value 3285.71
Step 4, added column 281, criterion value 3256.74
Step 5, added column 401, criterion value 3230.46
Step 6, added column 119, criterion value 3205
Step 7, added column 81, criterion value 3180.99
Step 8, added column 89, criterion value 3161.48
Step 9, added column 2, criterion value 3143
Step 10, added column 317, criterion value 3130.94
Step 11, added column 1696, criterion value 3120.39
Step 12, added column 325, criterion value 3109.96
Step 13, added column 448, criterion value 3100.45
Step 14, added column 42, criterion value 3091.67
Step 15, added column 1268, criterion value 3082.87
```

```
Step 16, added column 447, criterion value 3074.54
Step 17, added column 1458, criterion value 3066.79
Step 18, added column 505, criterion value 3058.97
Step 19, added column 285, criterion value 3050.64
Step 20, added column 1150, criterion value 3042.84
Step 21, added column 241, criterion value 3034.99
Step 22, added column 492, criterion value 3025.74
Step 23, added column 475, criterion value 3015.91
Step 24, added column 371, criterion value 3004.7
Step 25, added column 463, criterion value 2996.49
Step 26, added column 458, criterion value 2989.11
Step 27, added column 30, criterion value 2980.58
Step 28, added column 94, criterion value 2971.92
Step 29, added column 252, criterion value 2962.82
Step 30, added column 1609, criterion value 2955.1
Step 31, added column 1456, criterion value 2947.77
Step 32, added column 1558, criterion value 2940.63
Step 33, added column 1232, criterion value 2933.5
Step 34, added column 1537, criterion value 2926.61
Step 35, added column 1270, criterion value 2919.53
Step 36, added column 1796, criterion value 2912.79
Step 37, added column 145, criterion value 2906
Step 38, added column 519, criterion value 2897.2
Step 39, added column 105, criterion value 2889.69
Step 40, added column 92, criterion value 2882.67
Step 41, added column 536, criterion value 2875.48
Step 42, added column 556, criterion value 2868.86
Step 43, added column 1047, criterion value 2862.26
Step 44, added column 653, criterion value 2855.76
Step 45, added column 849, criterion value 2849.18
Step 46, added column 1821, criterion value 2842.79
Step 47, added column 1231, criterion value 2836.73
Step 48, added column 1435, criterion value 2830.68
Step 49, added column 1044, criterion value 2824.53
Step 50, added column 1024, criterion value 2818.74
Step 51, added column 7, criterion value 2813.13
Step 52, added column 15, criterion value 2798.9
Step 53, added column 45, criterion value 2792.79
Step 54, added column 165, criterion value 2787.09
Step 55, added column 418, criterion value 2780.41
Step 56, added column 1780, criterion value 2774.31
Step 57, added column 231, criterion value 2767.75
Step 58, added column 1121, criterion value 2762.05
Step 59, added column 1098, criterion value 2756.28
Step 60, added column 1380, criterion value 2750.57
Step 61, added column 1498, criterion value 2744.67
Step 62, added column 160, criterion value 2738.72
```

```
Step 63, added column 532, criterion value 2730.39
Step 64, added column 379, criterion value 2724.36
Step 65, added column 1235, criterion value 2718.68
Step 66, added column 1568, criterion value 2713.09
Step 67, added column 1203, criterion value 2707.25
Step 68, added column 1188, criterion value 2701.85
Step 69, added column 483, criterion value 2696.32
Step 70, added column 358, criterion value 2691.25
Step 71, added column 266, criterion value 2685.67
Step 72, added column 550, criterion value 2679.58
Step 73, added column 242, criterion value 2673.55
Step 74, added column 8, criterion value 2667.88
Step 75, added column 292, criterion value 2662.29
Step 76, added column 837, criterion value 2656.62
Step 77, added column 1759, criterion value 2651.08
Step 78, added column 1226, criterion value 2645.94
Step 79, added column 1058, criterion value 2639.39
Step 80, added column 1221, criterion value 2634.37
Step 81, added column 1311, criterion value 2629.34
Step 82, added column 301, criterion value 2624.59
Step 83, added column 393, criterion value 2618.85
Step 84, added column 1359, criterion value 2613.72
Step 85, added column 1643, criterion value 2609.14
Step 86, added column 1266, criterion value 2604.62
Step 87, added column 1349, criterion value 2599.01
Step 88, added column 988, criterion value 2594.48
Step 89, added column 1284, criterion value 2590.03
Step 90, added column 1013, criterion value 2585.57
Step 91, added column 900, criterion value 2581.13
Step 92, added column 748, criterion value 2576.04
Step 93, added column 800, criterion value 2570.4
Step 94, added column 765, criterion value 2565.49
Step 95, added column 661, criterion value 2560.5
Step 96, added column 702, criterion value 2556.02
Step 97, added column 614, criterion value 2549.17
Step 98, added column 1037, criterion value 2544.6
Step 99, added column 161, criterion value 2540.08
Step 100, added column 4, criterion value 2535.53
Step 101, added column 6, criterion value 2529.49
Step 102, added column 46, criterion value 2523.5
Step 103, added column 385, criterion value 2518.35
Step 104, added column 524, criterion value 2512.59
Step 105, added column 367, criterion value 2506.99
Step 106, added column 17, criterion value 2500.87
Step 107, added column 35, criterion value 2491.28
Step 108, added column 923, criterion value 2485.03
Step 109, added column 515, criterion value 2479.67
```

```
Step 110, added column 169, criterion value 2473.85
Step 111, added column 723, criterion value 2468.65
Step 112, added column 74, criterion value 2463.38
Step 113, added column 1595, criterion value 2458.03
Step 114, added column 66, criterion value 2452.71
Step 115, added column 409, criterion value 2447.56
Step 116, added column 1119, criterion value 2442.59
Step 117, added column 544, criterion value 2437.71
Step 118, added column 49, criterion value 2431.78
Step 119, added column 1823, criterion value 2426.6
Step 120, added column 61, criterion value 2422.05
Step 121, added column 26, criterion value 2416.71
Step 122, added column 573, criterion value 2411.54
Step 123, added column 613, criterion value 2406.29
Step 124, added column 68, criterion value 2401.31
Step 125, added column 567, criterion value 2396.66
Step 126, added column 1510, criterion value 2391.86
Step 127, added column 334, criterion value 2387.38
Step 128, added column 335, criterion value 2381.92
Step 129, added column 118, criterion value 2375.6
Step 130, added column 810, criterion value 2369.81
Step 131, added column 629, criterion value 2364.86
Step 132, added column 1427, criterion value 2360.08
Step 133, added column 1051, criterion value 2355.27
Step 134, added column 1224, criterion value 2350.53
Step 135, added column 1639, criterion value 2345.95
Step 136, added column 227, criterion value 2341.67
Step 137, added column 1466, criterion value 2337.4
Step 138, added column 987, criterion value 2332.77
Step 139, added column 1159, criterion value 2328.34
Step 140, added column 1492, criterion value 2324.1
Step 141, added column 1304, criterion value 2319.82
Step 142, added column 1181, criterion value 2315.67
Step 143, added column 1818, criterion value 2311.49
Step 144, added column 1641, criterion value 2307.19
Step 145, added column 926, criterion value 2303.15
Step 146, added column 510, criterion value 2299.03
Step 147, added column 216, criterion value 2294.47
Step 148, added column 1274, criterion value 2290.21
Step 149, added column 1312, criterion value 2286.05
Step 150, added column 1727, criterion value 2282.06
Step 151, added column 294, criterion value 2278.1
Step 152, added column 1073, criterion value 2274.02
Step 153, added column 1529, criterion value 2269.94
Step 154, added column 204, criterion value 2265.95
Step 155, added column 1068, criterion value 2261.93
Step 156, added column 1308, criterion value 2257.99
```

```
Step 157, added column 1036, criterion value 2254.08
Step 158, added column 1276, criterion value 2250.23
Step 159, added column 16, criterion value 2246.26
Step 160, added column 40, criterion value 2241.73
Step 161, added column 389, criterion value 2236.7
Step 162, added column 386, criterion value 2232.62
Step 163, added column 1392, criterion value 2228.64
Final columns included:  2 4 6 7 8 15 16 17 26 30 35 40 42 45 46 49 61 66
    68 74 81 89 92 94 105 118 119 145 160 161 165 169 204 216 227 231 241
    242 252 266 281 285 292 294 301 317 325 334 335 349 358 367 371 379 385
     386 389 393 401 409 418 447 448 458 463 475 483 492 505 510 515 519
    524 532 536 544 550 556 567 573 613 614 629 653 661 702 723 748 765 800
     810 825 837 849 900 923 926 987 988 1013 1024 1036 1037 1044 1047 1051
     1058 1068 1073 1098 1119 1121 1150 1159 1181 1188 1203 1221 1224 1226
    1231 1232 1235 1266 1268 1270 1274 1276 1284 1304 1308 1311 1312 1349
    1359 1380 1392 1427 1435 1456 1458 1466 1492 1498 1510 1529 1537 1558
    1568 1595 1609 1639 1641 1643 1696 1727 1759 1780 1796 1818 1821 1823
Number of features selected: 162
Selected feature indices: 2 4 6 7 8 15 16 17 26 30 35 40 42 45 46 49 61 66
     68 74 81 89 92 94 105 118 119 145 160 161 165 169 204 216 227 231 241
    242 252 266 281 285 292 294 301 317 325 334 335 349 358 367 371 379 385
     386 389 393 401 409 418 447 448 458 463 475 483 492 505 510 515 519
    524 532 536 544 550 556 567 573 613 614 629 653 661 702 723 748 765 800
     810 825 837 849 900 923 926 987 988 1013 1024 1036 1037 1044 1047 1051
     1058 1068 1073 1098 1119 1121 1150 1159 1181 1188 1203 1221 1224 1226
    1231 1232 1235 1266 1268 1270 1274 1276 1284 1304 1308 1311 131
Selected feature indices: 2 4 6 7 8 15 16 17 26 30 35 40 42 45 46 49 61 66
     68 74 81 89 92 94 105 118 119 145 160 161 165 169 204 216 227 231 241
    242 252 266 281 285 292 294 301 317 325 334 335 349 358 367 371 379 385
     386 389 393 401 409 418 447 448 458 463 475 483 492 505 510 515 519
    524 532 536 544 550 556 567 573 613 614 629 653 661 702 723 748 765 800
     810 825 837 849 900 923 926 987 988 1013 1024 1036 1037 1044 1047 1051
     1058 1068 1073 1098 1119 1121 1150 1159 1181 1188 1203 1221 1224 1226
    1231 1232 1235 1266 1268 1270 1274 1276 1284 1304 1308 1311 1312 1349
    1359 1380 1392 1427 1435 1456 1458 1466 1492 1498 1510 1529 1537 1558
    1568 1595 1609 1639 1641 1643 1696 1727 1759 1780 1796 1818 1821 1823
  From original predictors: 96 features (indices: 2 4 6 7 8 15 16 17 26 30
      35 40 42 45 46 49 61 66 68 74 81 89 92 94 105 118 119 145 160 161
     165 169 204 216 227 231 241 242 252 266 281 285 292 294 301 317 325
     334 335 349 358 367 371 379 385 386 389 393 401 409 418 447 448 458
     463 475 483 492 505 510 515 519 524 532 536 544 550 556 567 573 613
     614 629 653 661 702 723 748 7
  From original predictors: 96 features (indices: 2 4 6 7 8 15 16 17 26 30
      35 40 42 45 46 49 61 66 68 74 81 89 92 94 105 118 119 145 160 161
     165 169 204 216 227 231 241 242 252 266 281 285 292 294 301 317 325
     334 335 349 358 367 371 379 385 386 389 393 401 409 418 447 448 458
     463 475 483 492 505 510 515 519 524 532 536 544 550 556 567 573 613
```

```
     614 629 653 661 702 723 748 765 800 810 825 837 849 900 923 )
  From change rate predictors: 66 features (indices: 3 64 65 90 101 113
     114 121 124 128 135 145 150 175 196 198 227 236 258 265 280 298 301
     303 308 309 312 343 345 347 351 353 361 381 385
  From change rate predictors: 66 features (indices: 3 64 65 90 101 113
     114 121 124 128 135 145 150 175 196 198 227 236 258 265 280 298 301
     303 308 309 312 343 345 347 351 353 361 381 385 388 389 426 436 457
     469 504 512 533 535 543 569 575 587 606 614 635 645 672 686 716 718
     720 773 804 836 857 873 895 898 900 )

=== AUC vs Number of Top Features ===
Top K      Train AUC       Test AUC
----------------------------------------
5          0.9045          0.8890
10         0.9047          0.8889
15         0.9057          0.8878
20         0.9063          0.8873
25         0.9076          0.8837
30         0.9080          0.8841
50         0.9093          0.8838

Done.
```

Listing 4: Results.txt — Full MATLAB console output